

Doc. 308 – The Cosmic Sector of FFGFT:  
Redshift, Dark Sector and  
Geometric Gravity

Johann Pascher

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# Doc. 308 – The Cosmic Sector of FFGFT

## What this is about

FFGFT advances an unusual claim: the universe is not expanding. Cosmic redshift is not a Doppler effect but a geometric property of the  $\xi$ -field. Dark energy does not exist as a physical entity. And dark matter requires no particles — its effects are to follow from the same geometry that carries light deflection; this is established for gas-dominated dwarf galaxies, for spiral galaxies and the Bullet Cluster it is open.

This document assembles the cosmic sector of FFGFT: what has been derived, what has been observationally confirmed, and where the limits lie. It addresses readers with a physics background who want to know *what* the theory says and *why*.

The three sources of degeneracy are named, the two cores of the geometry are developed, and the status of each step — derivation, structural argument, posit — is declared.

## Conventions

### Legend.

- [K] *Core derivation* — derived mathematically or numerically from the foundational posits; reproducible.
- [S] *Structural claim* — justified but not yet fully proved.
- [POSIT] *Posit* — declared assumption, not further derived.

### Central quantities.

|                                            |                                                                                                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\xi = 4/30000$                            | Only fundamental parameter of FFGFT [POSIT, P33].                                                                                                               |
| $\tilde{T} \cdot m = 1$                    | Foundational relation: intrinsic time and mass are inversely coupled.                                                                                           |
| $R_H = c/H_0$                              | Geometric maximal scale — not an expansion horizon.                                                                                                             |
| $a_0 = c^2 \xi^{10} / (4 \bar{\lambda}_e)$ | Transition scale of the inertia regime; below $a_0$ inertia is non-linear ( $\tilde{T} \cdot m = 1 + \text{Unruh}$ ); $= 1.033 \times 10^{-10} \text{ m/s}^2$ . |
| $\bar{\lambda}_e = \hbar / (m_e c)$        | Reduced Compton wavelength of the electron.                                                                                                                     |

# Chapter A

## The Static Universe and Redshift

### A.1 The fundamental choice

$\Lambda$ CDM describes an expanding universe: galaxies move apart, light is redshifted by the expansion of space, and the observation of distant supernovae reveals an accelerated expansion accounted for by a cosmological constant  $\Lambda$ .

FFGFT makes a different fundamental choice: the universe is static. Cosmic redshift arises not from motion but from geometric path elongation in the  $\xi$ -field along the light path.

### A.2 The redshift mechanism [K]

The FFGFT redshift formula (Doc. 280):

$$1 + z = e^{\xi x} \quad \implies \quad z \approx \xi x \quad (\text{for small } z),$$

where  $x$  is the path length, measured in a length that is not derived from  $\xi$  and is fixed only via the measured  $H_0$ : the form of the relation is derived, its scale is calibrated (P39). Each photon loses energy along its path through fractal path elongation — proportional to distance, without wavelength dependence.

**Achromatic [K]:** The path elongation is purely geometric and stretches *all* wavelengths by the same factor  $e^{\xi x}$ . Finite-element simulations of the geometry confirm this (K6, Doc. 190). A chromatic redshift would destroy the coherence of the CMB spectral shape — this is not observed.

### A.3 The degeneracy of the data [K]

For the three classical pillars of cosmology — Type Ia supernovae, baryon acoustic oscillations (BAO), CMB temperature — the following holds: they cannot in principle distinguish between expansion and static geometric path stretching, as long as both models produce the same distance-redshift relation  $z(d)$  (Doc. 267).

This is not a weakness of FFGFT. It is a methodological statement: *these data alone do not decide between the models*. The decision shifts to theoretical consistency, parameter count and derivability.

## A.4 What follows from this

**For dark energy:**  $\Lambda$  is needed in  $\Lambda$ CDM to account for the apparent accelerated expansion. In the static reading there is no expansion to be accelerated.  $\Lambda$  loses its referent in the static reading — it is not falsified, it becomes objectless.

**For the Hubble tension:** Different measurement methods for  $H_0$  give different values. In the FFGFT reading,  $H_0$  does not measure an expansion rate but a parameter of the  $\xi$ -geometry. FFGFT knows only *one*, calibrated  $H_0$  value and does not distinguish between early and late measurements; the tension cannot be modelled in FFGFT — it is neither reproduced nor resolved.

**For the  $z$ -axis:** Every presentation of cosmological data “over time” is formulated in  $\Lambda$ CDM language. An FFGFT interpretation must treat  $z$  as a geometric distance parameter, not a time axis.

## Chapter B

# Geometric Gravity: the Two-Halves Synthesis

### B.1 Einstein 1911 and 1915

Einstein calculated the deflection of light at the Sun in 1911 from time dilation alone:

$$\alpha_{1911} = \frac{2GM}{c^2 b} = 0.875''.$$

That was half the measured value. The 1915 theory of GR yielded the full value  $\alpha_{1915} = 1.750''$ , because space *and* time curvature each contribute one half.

FFGFT *explains* this factor of 2 from first structure.

### B.2 K3 area invariance [K]

$\tilde{T} \cdot m = 1$  (time curvature) and fractal path elongation (space curvature, Doc. 182) are in FFGFT not alternative mechanisms but **complementary halves**. The K3 rule — area invariance of the time-space 2-cell — anchors this:

$$L_t \cdot l = \text{const.}$$

This is formally the reciprocity form of  $\tilde{T} \cdot m = 1$ , applied to the time-space pair.

**FE verification [K]:** Ray tracing through three configurations (30001 log nodes, RK4, `ffgft_308_p36_stufeB1_faktor2_fe.py`):

| Configuration                              | $\alpha$ at solar limb | Ratio to 1.75'' |
|--------------------------------------------|------------------------|-----------------|
| Time rate only ( $\tilde{T} \cdot m = 1$ ) | 0.875''                | 0.500           |
| Path elongation only (space)               | 0.875''                | 0.500           |
| Both together (K3)                         | 1.750''                | 1.000           |

**PPN test [K]:** K3 is the sole survivor of the  $T^4$  selection computation (four candidates tested, `ffgft_308_p36_stufeB2_kappa_herleitung.py`):  $\gamma_{\text{PPN}} = 1$  exactly. Cassini bound  $|\gamma - 1| < 2.3 \times 10^{-5}$  passed with unlimited margin.

### B.3 The same structure in the inertia regime: $a_0$ [K]/[S]

The K3 two-halves structure appears a second time, at cosmological scale. From Doc. 279 (the exponent 10 is calibrated to  $H_0$ , P20/P39; Doc. 309):

$$\frac{H_0}{c} = \frac{\pi}{2} \frac{\xi^{10}}{\bar{\lambda}_e} \quad \Rightarrow \quad R_H = \frac{2 \bar{\lambda}_e}{\pi \xi^{10}}.$$

$R_H$  is the geometric maximal scale of the compact  $T^4$  geometry (Doc. 182) — not an expansion horizon, valid in the static reading.

**The cycle doubling [S]:** The K3 2-cell has two cycles. A closed path that traverses each once measures:

$$L = \underbrace{2\pi R_H}_{\text{time cycle}} + \underbrace{2\pi R_H}_{\text{space cycle}} = 4\pi R_H.$$

**Rindler saturation [S]:** The Rindler horizon of an observer with acceleration  $a$  lies at  $c^2/a$ . When it reaches  $L$ , the acceleration is no longer relationally definable. This sets:

$$a_{bg} = \frac{c^2}{4\pi R_H} = \frac{c^2 \xi^{10}}{8 \bar{\lambda}_e}.$$

**Inertia from  $\tilde{T} \cdot m = 1$  [S]:** The foundational relation is a statement about mass. If inertia follows the time field (Unruh coupling), the force law in closed form is:

$$\boxed{a = \sqrt{a_N^2 + a_N a_0}} \quad \text{with} \quad a_0 = 2 a_{bg} = \frac{c^2 \xi^{10}}{4 \bar{\lambda}_e} = 1.033 \times 10^{-10} \text{ m/s}^2.$$

The  $2\pi$  of the geometry cancels exactly against the  $\pi/2$  from Doc. 279. The rational factor 4 remains:

$$4 = \underbrace{2}_{\text{K3: time and space cycle}} \times \underbrace{2}_{\text{Unruh: } a_0=2a_{bg}}.$$

No step contains a further free parameter; the scale takes over the calibration of the exponent 10 to  $H_0$ .

# Chapter C

## The Dark Sector

### C.1 Dark energy: not a physical object

$\Lambda$  arises in  $\Lambda$ CDM as an answer to the apparent accelerated expansion. In the static reading this expansion does not exist.  $\Lambda$  thereby loses its referent — it is not falsified by the reading change, it becomes objectless.

Three conditions speak against it:

1. **Only from degenerate data.** SNe Ia, BAO, CMB — exactly the datasets between which both readings are indistinguishable.
2. **No independent anchor.** The only theoretical prediction (QFT vacuum energy) misses the value by  $10^{122}$ .
3. **Objectless under reading change.** Not refuted, but without referent.

That is the status: reading artefact. Not “dark energy does not exist” — that would be a statement within the expansion reading — but: it is a reading-internal bookkeeping quantity.

### C.2 Dark matter: a regime, not a substance

The findings — rotation curves, gravitational lensing, Bullet Cluster — are reading-independent. They do not disappear under a reading change. FFGFT traces them, without a new ingredient, to the same force law that  $\hat{T} \cdot m = 1$  together with K3 geometry delivers; this is established for the gas-dominated dwarf galaxy DDO 154, for spiral galaxies and the Bullet Cluster it is open (see below).

Dark matter in FFGFT is not a substance. It is a name for a regime.

**The force law in the galactic regime [K]/[S]:**

$$a = \sqrt{a_N^2 + a_0 a_N}.$$

For  $a_N \gg a_0$  (inner regions):  $a \rightarrow a_N$  (Newton). For  $a_N \ll a_0$  (outer regions):  $a \rightarrow \sqrt{a_N a_0}$ , hence  $v^4 = GM_b a_0$  — the baryonic Tully-Fisher relation.

**In FFGFT, dark matter does not denote a missing mass component but the regime below  $a_0$  in which inertia is non-linear. The missing mass is not missing — inertia is different there.**

**Results [K] (ffgft\_308\_p36\_stufeA\_a0\_anker.py):**

| Galaxy    | Type                 | $v_{\text{Newton}}$ | $v_{\text{FFGFT}}$ | $v_{\text{obs}}$ |
|-----------|----------------------|---------------------|--------------------|------------------|
| DDO 154   | dwarf, gas-dominated | 18.5                | 47.1               | 47.0 km/s        |
| NGC 3198  | spiral               | 50.4                | 121                | 150 km/s         |
| Milky Way | spiral               | 122                 | 180                | 220 km/s         |



DDO 154 is the cleanest test (gas-dominated,  $M_b$  best known):  $v_{\text{FFGFT}}/v_{\text{obs}} = 1.002$ . The deviations for NGC 3198 and the Milky Way ( $121/150 - 1 = -19.3\%$ ,  $180/220 - 1 = -18.2\%$ ) lie well outside the  $M_b$  uncertainty: for  $v \propto M_b^{1/4}$ ,  $\pm 30\%$  in  $M_b$  correspond to only  $+6.8\% / -8.5\%$  in  $v$  (even in the Newtonian limit only  $+14\%$ ). Closing them would require  $M_b \times 2.36$  (NGC 3198) and  $\times 2.23$  (Milky Way). For spirals the test is open.

**Bullet Cluster [K] (`ffgft_308_p36_stufe01_bullet_cluster.py`):** Two-clump FE projection (Clowe+2006 parameters) at instantaneous coupling. According to Clowe et al. 2006 the X-ray gas is the dominant baryonic component ( $M_{\text{gas}}/M_{\text{stars}} \approx 5-10$ ; gas  $\sim 85\%$  of the baryons, i.e.  $5.7 : 1$ ). With these baryonic masses the lensing peak lies at 240 kpc, i.e. at the gas; it moves to the gas already from gas:stars  $\approx 1.4$ . A galaxy mass of  $2.3 \cdot 10^{14} M_{\odot}$  would be the lensing mass including the missing component; a test with it is circular. The Bullet Cluster is therefore not established as passed; the test is open.

### C.3 Asymmetry in comparison

$\xi$  is cross-anchored — it carries the lepton mass ladder, the Koide scalar, the  $\alpha$  reference,  $K_{\text{frak}}$  and  $k^*$  (Doc. 275). It works in five foreign sectors before the cosmological one.

$\Delta$  works nowhere except its own fit.

That is the difference: not "true vs. false" but "cross-anchored vs. isolated".

## Chapter D

# Summary: Picture of the Cosmic Sector

| Quantity                               | FFGFT                                                   | Status            |
|----------------------------------------|---------------------------------------------------------|-------------------|
| Cosmic redshift                        | $1 + z = e^{\xi x}$ , scale via $H_0$                   | [K] (form)        |
| Achromatic                             | geometric, FE confirmed                                 | [K]               |
| Degeneracy SNe/BAO/CMB                 | between both readings                                   | [K]               |
| $\Lambda$ (DE)                         | reading artefact, no referent                           | [K]               |
| Hubble $H_0$                           | $c\pi\xi^{10}/(2\bar{\lambda}_e)$ , exponent calibrated | [POSIT]           |
| Light deflection                       | $0.875'' + 0.875'' = 1.750''$                           | [K]               |
| $\gamma_{\text{PPN}} = 1$              | K3 sole survivor                                        | [K]               |
| $a_0 = c^2\xi^{10}/(4\bar{\lambda}_e)$ | chain via calibrated $H_0$                              | [K]               |
| Rotation curves                        | DDO 154 ratio 1.002; spirals ~18 % low                  | [K]; spirals open |
| Bullet Cluster                         | baryonic: peak at the gas (240 kpc)                     | open              |
| K3 cycles $4\pi R_H$                   | time + space cycle equal                                | [S]               |
| Rindler saturation                     | horizon meets maximal scale                             | [S]               |
| $\xi = 4/30000$                        | magnitude                                               | [POSIT]           |
| K3 forward proof                       | 2-cell vs. 4-volume                                     | open              |
| 14% to $a_0^{\text{emp}}$              | literature range $1.1\text{--}1.3 \times 10^{-10}$      | open              |

The cosmic sector of FFGFT rests on a single mechanism:  $\tilde{T} \cdot m = 1$  together with K3 area invariance produces light deflection, geometric gravity and achromatic redshift.  $a_0$  follows from the same chain without a further free parameter; the scale carries the calibration of  $H_0$ .

Dark energy is not a physical object — it is a construct of the expansion reading. Dark matter as particles is not required insofar as the effects follow from the geometry — established for gas-dominated dwarf galaxies, open for spiral galaxies and the Bullet Cluster.

# Chapter E

## References

**Internal documents:** Doc. 025 (cosmology overview); Doc. 026 (FE geometry); Doc. 064 (redshift foundations); Doc. 182 (maximal scale); Doc. 267 (cosmological degeneracy); Doc. 279 ( $H_0$  from  $\xi^{10}$  chain, K6); Doc. 280 (achromatic redshift); Doc. 275 ( $k^*$  recursion); Doc. 190 (P33, P44, K6).

**Verification scripts (all inputs declared):** `ffgft_308_p36_stufeA_a0_anker.py` (chain  $\xi \rightarrow a_0$ , rotation curves); `ffgft_308_p36_stufeB1_faktor2_fe.py` (factor-2 gate, 30001 log nodes, RK4); `ffgft_308_p36_stufeB2_kappa_herleitung.py` (selection computation K3); `ffgft_308_p36_stufeC1_bullet_cluster.py` (Bullet Cluster projection).

**External references:** McGaugh, Lelli, Schombert 2016, PRL 117, 201101 (RAR/ $a_0$ ); Clowe et al. 2006, ApJ 648, L109 (Bullet Cluster).